# Lexical Task Heads on Simple Tasks

The best approaches so far, compared

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## Setting and how to read the numbers

Llama-3.1-8B-Instruct (bf16), the 14 simple tasks of Todd et al. (antonym, capitals, translations, ...), rule style unless stated. A head set is tested by patching its mean output ($\times\alpha$) at the last prompt token:

- **Check 1**: zero-shot prompts, which do not say what the task is (unpatched accuracy $\approx 0$). The main test: does the set *bring in* the task?
- **Check 2**: 1-shot prompts the model answered wrongly (baseline 0). The prompt already shows the task, so this check tests repair, not task transfer.

Readouts: **Logit-head** (head write decoded by the unembedding, as Yang et al.) and **J-head** (Head-JLens: head output through its fitted Jacobian, then decoded). Accuracy is the flexible score. Unless stated, a number is the **mean over the 14 tasks in check 1**.

The numbers come from three protocols, marked in Table 1: **held-out** (E4: heads, means and $\alpha$ chosen on one half of the items, scored on the other; the fairest), **same items** (chosen and scored on the same items, best $\alpha$: optimistic), and **one fold** (held-out items, but best $\alpha$ in fold 1 only). Every run is single, greedy, without intervals.

## 1 Starting point A: lexical task heads (Yang et al., 2026)

**Their method.** A head is a lexical task head if, on $\geq$10% of 100 correctly answered prompts, one of its 10 top decoded tokens (logit lens) is in a hand-written term list for the task (2–22 tokens, e.g. *opposite*, *capital*). The set's mean output is then patched into prompts without task information.

**What we changed.**

1. *Padding masked.* Their code batches prompts padded on the left without an attention mask, so the model attends to the padding. We mask it: each prompt is computed as if run alone.
2. *J-head* as a second readout, beside their Logit-head.
3. *Scoring.* Their first-token match marks any number correct and capitalised German nouns wrong; we generate the reply and use a flexible score (whole answer as words, case/accents/articles ignored). Heads, means and patched items are all chosen by it.
4. *Second check.* The paper's zero-shot figures (App. G.1) are reproduced only by the setting of their one released script: 1-shot prompts answered wrongly. We report both checks.
5. bf16 instead of fp32 (the precision the lens was fitted in).

**How it worked.**

- $\sim$ *Replication.* Head counts within one head of Yang's code-setting figures. Check 1 works only on fact-like tasks (capitals, landmark, park, product); nothing on antonym, synonym, present–past, prev-item and translations. Check 2 gains on all 14 and matches their figures.
- ✓ *J-head.* Finds more heads (on 23/28 task $\times$ style pairs), the extra ones in layers 10–19; the heads only J-head finds have effects of their own; its lead holds at equal set size (16/25 leads at 5, 10 and 20 heads, none reversed). It does not contain the Logit-head set (H2 falsified).
- ✗ *Which templates.* Heads that pass under all 5 templates help only fact tasks; restricting the union to heads shared across tasks removes its gains.
- **Held-out:** the E1 J-head set reaches **0.20**. It does nothing on antonym, singular–plural, present–past and the three translations ($\leq$0.03).

## 2 Starting point B: instruction function vector (Davidson et al., 2025)

**Their method.** Each head's causal score: replace the prompt's instruction by uninformative text (equiprobable tokens, WikiText, other tasks' instructions; matched in length and log-probability), set the head to its mean output, and measure the gain in the answer's probability. The 20 heads with the highest score averaged over tasks form one set; the function vector (FV) is the sum of their mean outputs, added once after layer $|L|/3$ (layer 11) at the last token, $\alpha = 1$.

**What we changed.**

1. *Instructions and scoring.* Our rule templates inst1–4 as the instructions (25 correct prompts each for the mean), flexible score.
2. *Test.* Our check 1 with its control and an $\alpha$ grid, instead of their evaluation.
3. *Set size and scope.* 4 tasks first, then heads chosen over all 14 tasks (30 heads); a 5–1,024 size sweep.
4. *Per-task ranking, held out* (E4): the first $N$ heads by each task's own score, computed on the choosing half: FV-20, FV-40, FV-$N_t$ (as many heads as CC-pooled, below).
5. *Generic heads removed* (FV split): 6 of the 30 heads decode the same tokens on every task (line ends, colons), attend to the answer slot, and give one constant output. FV minus these 6 = *task24*.

**How it worked.**

- ✓ *Their 30-head FV* at $\alpha = 1$: **0.39** on held-out items, twice E1 (0.20); in its first run (same items) $\geq$ our best E1 set on 11/14 tasks. Nothing at $|\alpha| \geq 3$.
- ✓ *Per-task ranking*: FV-$N_t$ **0.59**, the best held-out set; also best at 20 (0.38) and 40 heads (0.43). It alone carries present–past (0.85). It fails on English–German (0.01) and is below CC-pooled on all three translations and prev-item.
- ✓ *Generic heads removed*: task24 **0.58** vs the full FV 0.33 on the same items, $\geq$ it on all 14 tasks; translations 0.39–0.70 (FV $\leq$0.10). The 6 generic heads are most of the FV's norm, in one direction shared by all tasks; scaled up alone, they turn replies into *://* or *1. 2. 3.*; without them the FV works best at $\alpha$ 2–3 instead of 1. Removing 6 random heads does not help.
- ∼ *Reading the FV heads*: under J-head about 20 of 30 are readable (answer category: L16H25 *cities*, L17H4; domain; relation: L13H22 *opposite, next, earlier*). Only 3–11 per task are in our decoded-word sets. A lens-only estimate of their score finds 7–8 of the 30 (late layers), not the mid-layer word heads.

## 3 What we built on them

**CC-pooled: term lists from decoded words** (Yang's criterion, our lists; J-head). Instead of the hand-written list, 5 *task words* (decoded on this task more than on the other 13) + 5 *common words* (decoded on every task: *what, which, maybe, https*), with words and heads counted on the 5 templates pooled. 95–154 heads. ✓ Held-out **0.44** vs E1 0.20 and layer-matched random sets 0.02; above E1 on 13/14. Carries the tasks E1 cannot (antonym, singular–plural, translations 0.38–0.40). Fails on present–past (0.02). At 20 and 40 heads it falls to 0.22 / 0.26, below FV at the same size. Both word kinds and the pooling are needed; removing WikiText-generic words helps facts and translations but hurts antonym, synonym and sequences.

**Attend + decode** (Yang's readout, ranking by mechanism). Heads that attend to the instruction's task words and decode a task word, after L16H25 (attends *capital*, *country*; decodes *city*). ∼ Few heads carry named answer types: capitals 0.88 with L16H25 alone, landmark 0.79 (5 heads), product–company 0.72–0.80 (20 heads). Nothing on translations, sequences, present–past; mean 0.25 at 20 heads (0.28, one fold).

**LLM-scored heads** (E5; GPT-5.5 reads each head's top-20 decoded tokens and scores it 0–100 for representing the *task*, not the answer or the input). Run on 2 tasks only, held-out. ∼ Country–capital: J-head 20 heads 0.78, Logit-head 0.70, ≈ CC-20 0.77, below FV-20 0.88. English–German: nothing in check 1 below 60 heads, then J-head 0.21–0.26 vs full CC 0.39; Logit-head sets 0 at every size. J-head sets beat Logit-head sets on English–German at every size and on country–capital at 20–30 heads (not at 40: 0.59 vs 0.68).

**Single heads that carry a task.** L16H25 for capitals (above). L15H1 for present–past: with example-style means it alone gives 0.80 (check 1) and 0.76 (check 2), control 0, while the 4-head threshold set gives 0; J-head reads past participles from it, Logit-head reads nothing.

# 4 Side by side

| Head set | Origin | Heads | Check 1 | Check 2 | Protocol |
|---|---|---|---|---|---|
| E1, J-head | Yang + J-head | 7–50 | 0.20 | 0.31 | held-out |
| CC-pooled | Yang, our lists | 97–154 | 0.44 | 0.33 | held-out |
| CC-20 / CC-40 | Yang, our lists | 20 / 40 | 0.22 / 0.26 | 0.28 / 0.34 | held-out |
| FV-30 ($\alpha = 1$) | Davidson, as published | 30 | 0.39 | – | held-out |
| FV-20 / FV-40 | Davidson, per task | 20 / 40 | 0.38 / 0.43 | 0.33 / 0.39 | held-out |
| **FV-$N_t$** | Davidson, per task | 97–154 | **0.59** | 0.39 | held-out |
| Random-$N_t$ | control | 97–154 | 0.02 | 0.12 | held-out |
| FV-univ-20 | Davidson, over 14 tasks | 20 | 0.63 | 0.39 | one fold |
| Attend + decode-20 | Yang, by mechanism | 20 | 0.28 | 0.35 | one fold |
| **task24** | Davidson minus generic | 24 | **0.58** | – | same items |
| FV-30, same run | Davidson, as published | 30 | 0.33 | – | same items |

Table 1: Mean accuracy over the 14 tasks. Held-out: cross-fitted $\alpha$, mean of the two folds. One fold: best $\alpha$ in fold 1. Same items: best $\alpha$ on the items scored (task24 at a fixed $\alpha = 3$: 0.54). Rows of different protocols are not directly comparable. –: not run in check 2. LLM-scored sets (2 tasks only) are not in the table.

**By task type** (held-out, check 1):

- *Fact tasks* (capitals, landmark, park, product): every method works; FV-$N_t$ 0.88–0.94; a single head can be enough (L16H25).
- *Translations and prev-item*: only decoded-word sets: CC-pooled 0.38–0.40 vs FV-$N_t$ 0.01–0.25. task24 (same items) also carries translations.
- *Present–past*: only the causal ranking (FV-$N_t$ 0.85), or L15H1 with example means. No decoded-word set with rule means.
- *Antonym, synonym, singular–plural, next-item*: FV-$N_t$ first (0.49–0.85), CC-pooled second (0.26–0.69), E1 near 0.

# 5 What did not help

- ✗ Heads passing under all 5 templates ($k = 5$), and strict consensus lists: better on fact tasks only.
- ✗ Union restricted to heads shared across tasks: loses the union's gains.
- ✗ Heads from what the instruction adds (E6, clean − corrupted readings): 0.18 vs CC 0.44; the heads whose words do not depend on the instruction are needed.
- ✗ Word lists without the cross-task contrast (task minus WikiText, 0.21) or with a within-group contrast (0.32).

# 6 Main insights

1. **J-head is the better readout** for finding task heads: more heads, earlier layers, causal effects that remain at equal size; LLM-scored J-head sets also mostly beat Logit-head ones.
2. **The hand-written term list is the bottleneck of Yang's method**, not the criterion: lists built from what heads actually decode double the held-out effect (0.20 → 0.44).
3. **A causal ranking still finds better heads** than any decoding rule (FV-$N_t$ 0.59), except on translations and prev-item, where decoded words win.
4. **Davidson's FV is held back by a few generic heads** that add a large, task-independent direction; without them it works at larger $\alpha$ and also carries translations.
5. **Check 2 does not separate the sets** (all sets 0.28–0.39, random sets 0.12, control means up to 0.13): the prompt already gives the task. Check 1 is the informative test.
6. **Read the replies.** 19–33% of patched replies on several tasks contain the answer after a prefix the score rejects; part of a "loss" can be a format change.

**Open:** task24 held-out (only same-items so far); is the generic FV part harmful through its heads or only its norm; LLM-scored heads on the other 12 tasks; E4b (new wordings, held-out wordings) not run.